

Semigroups — Zero to Hero — Worksheet

James Mitchell¹, Reinis Cirpons², Joseph Edwards¹, and Finn Smith¹

¹University of St Andrews

²Nantes Université, École Centrale Nantes, CNRS, Inria, LS2N, UMR 6004, France

May 2026

If you've never used GAP before you might want to look at:

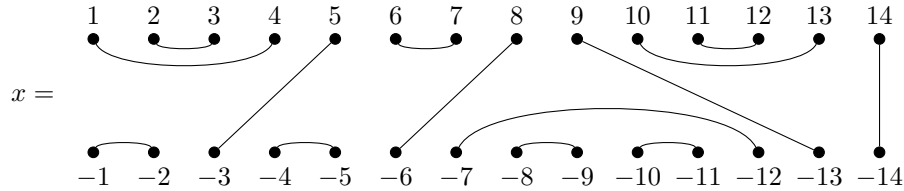
<https://carpentries-incubator.github.io/gap-lesson/>

before attempting any of the problems on this sheet.

This worksheet contains some exercises to be completed in GAP with the use of the **Semigroups** package as part of the GAP workshop session of NBSAN 40. The problems have been written in a deliberately terse way to encourage attendees to experiment with the package and explore its associated manual: https://semigroups.github.io/Semigroups/doc/chap0_mj.html.

As well as inspecting the package manual, please also feel free to ask questions to James, Reinis or Joe!

1. Let J_n denote the Jones monoid of degree n , and



be a bipartition of degree 14.

- (a) Find a collection of bipartitions $\{x_1, x_2, \dots, x_j\}$ such that the tensor product $x_1 \otimes x_2 \otimes \dots \otimes x_j$ is equal to x . What is the largest possible value of j ? For the definition of the tensor product, see the documentation of the function **TensorBipartition**.
- (b) How many elements does J_{14} contain?
- (c) How many idempotent elements does J_{14} contain?
- (d) How many idempotent elements does the Green's $\mathcal{D}$ -class of x in J_{14} contain?
- (e) How many idempotent elements e , for which 7 is in the domain of e , does the Green's $\mathcal{D}$ -class of x in J_{14} contain?
- (f) How many irreducible idempotent elements $\hat{e}$, for which 7 is in the domain of $\hat{e}$, does the Green's $\mathcal{D}$ -class of x in J_{14} contain?

2. Recall that the commuting graph of a semigroup S is the graph with nodes the elements of S and an edge (x, y) if $xy = yx$ holds. Let T_n be the full transformation monoid on $n \leq 5$ points. Show that the clique numbers of the commuting graph of T_n are 2^{n-1} .

3. Let S be the semigroup defined by the presentation

$$\langle a, b, c, d, e, f, g \mid abcd = a^3ea^2, ef = dg \rangle.$$

- (a) Show that S is infinite.
 - (b) Create a homomorphism from the free semigroup $\{a, b, c, d, e, f, g\}^+$ to S .
 - (c) Partition the first 1000 elements of $\{a, b, c, d, e, f, g\}^+$ so that words belong to the same part if and only if they represent the same element of S .
4. Is the monoid defined by the presentation

$$\langle x_2, \dots, x_n \mid x_i^2 = (x_i x_j)^3 = (x_i x_j x_k)^4 = 1 \quad i, j, k \text{ distinct} \rangle$$

the symmetric group for $n \geq 2$?

5. Determine which of the relations in the presentation are redundant and which are not:

$$\langle a, b \mid a^4 = 1, b^2 = b, ba^3ba = a^2(ab)^2, (ba^2)^2 = (a^2b)^2, (ba)^2a^2 = aba^3b, a(ab)^4 = (ab)^4 \rangle.$$

What is the minimal set of the relations in this presentation that define the same monoid?

6. Let S be the Catalan monoid of degree 3.

- (a) Draw the egg-box diagram of S .
 - (b) Draw the left and right Cayley graphs of S .
 - (c) Show that S has two non-trivial non-universal non-Rees congruences.
 - (d) Show that the lattice of left and right congruences of S are isomorphic.
 - (e) What are the maximal subsemigroups of S ? Show that none of the maximal subsemigroup is isomorphic to any of the others.
7. Let S be the dual of the full transformation monoid on 5 points. Find a transformation representation of S on 32 points.
- (a) Draw the egg-box diagram of S .
 - (b) Draw the left and right Cayley graphs of S .
 - (c) Show that S has two non-trivial non-universal non-Rees congruences.
 - (d) Show that the lattice of left and right congruences of S are isomorphic.
8. The translational hull $\Omega(S)$ of a semigroup S is the set of all bitranslations (λ, ρ) under componentwise composition. Any element s of S induces an inner bitranslation (λ_s, ρ_s) of S , where $\lambda_s(x) = sx$ and $\rho_s(x) = xs$ for all $x \in S$. This gives a homomorphism π_S from S to $\Omega(S)$. If this is injective, then S is *weakly reductive*. Restricting the bitranslations to an ideal I of S gives a homomorphism $\pi_{S|I}$ from S to $\Omega(I)$. If I is weakly reductive, then I is densely embedded in S if and only if $\pi_{S|I}$ is an isomorphism.

- (a) Determine the size of $\Omega(I)$ for each ideal I of J_7 (the Jones monoid of degree 7).
- (b) Which of the ideals of J_7 are weakly reductive?
- (c) Which of the ideals of J_7 are densely embedded in J_7 ?
- (d) One such $\Omega(I)$ is gigantic, and so is certainly not being enumerated. Using `KnownTruePropertiesOfObject`, conjecture how the size is being calculated.